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SUBSEA COMPLETION SYSTEMS

Abstract

A subsea completion system can include a lower completion assembly (LCA) having a receptacle disposed at a top end of the LCA, where the LCA comprises a LCA wall that forms a LCA cavity; a first barrier disposed within the LCA cavity, where the first barrier forms a first fluidic seal with the LCA wall, and where the first barrier is configured to open when a first differential force within the LCA cavity reaches a first threshold value; an upper completion assembly (UCA) having a tubing string that includes a tubing string wall that forms a tubing string cavity; and a second barrier disposed within the tubing string cavity, where the second barrier forms a second fluidic seal with the tubing string wall, and where the second barrier is configured to open when a second differential force within the tubing string cavity reaches a second threshold value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/562,497 titled "Subsea Completion Systems" and filed on Mar. 7, 2024, the entire contents of which are hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] The present application is related to subsea field operations and, more particularly, to subsea completion systems.

BACKGROUND

[0003] During subsea completions using a horizontal Xmas tree system, the final step of installing barriers into the well is to use slickline to install crown plugs in the tubing hanger. This process can take 3 to 5 days of around-the-clock operation, cost large amounts of money in equipment rental, and pose a number of significant safety concerns. For example, in order to install crown plugs in the tubing hanger, an operator would need to rig up a compensated coiled tubing lift frame (CCTLF), rig up a surface flow tree (SFT) tool, rig up a slickline, run the slickline to retrieve the isolation sleeve, install the crown plugs, rig down the slickline, rig down the SFT, and rig down the CCTLF.

[0004] In the current art, running crown plugs in place cannot be done because there is no way to prevent wellbore packer fluid from entering the tubing string on installation. Packer fluid is damaging to the formation, and there is no way to properly circulate fluids to eliminate packer fluid from being interfaced with the well. In addition, running crown plugs in place cannot be done because the production bore of the tubing hanger is open (not sealed over). If the well begins to flow prior to the tubing hanger engaging the Xmas tree, there is no way to shut in the BOP and stop the flow without damaging the completion as the open port and tubing would provide a conduit out of the well above the BOP.

SUMMARY

[0005] In general, in one aspect, the disclosure relates to a subsea completion system that includes a lower completion assembly (LCA) having a receptacle disposed at a top end of the LCA, where the LCA includes a LCA wall that forms a LCA cavity. The subsea completion system may also include a first barrier disposed within the LCA cavity, where the first barrier forms a first fluidic seal with the LCA wall, and where the first barrier is configured to open the first fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value. The subsea completion system may further include an upper completion assembly (UCA) having a tubing string disposed toward a bottom end of the UCA, where the tubing string includes a tubing string wall that forms a tubing string cavity. The subsea completion system may also include a second barrier disposed within the tubing string cavity, where the second barrier forms a second fluidic seal with the tubing string wall, and where the second barrier is configured to open the second fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value.

[0006] In another aspect, the disclosure relates to a method of establishing a subsea completion system. The method can include inserting a first barrier within a top end of a lower completion assembly (LCA) cavity formed by a LCA wall of the LCA, where first barrier forms a first fluidic seal with the LCA wall, where the first barrier is configured to open the first fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and where the LCA with the first barrier is configured to be subsequently inserted into a casing string. The method can also include inserting a second barrier toward a bottom end of a tubing string of an upper completion assembly (UCA), where the tubing string has a tubing string cavity formed by a tubing string wall, where the second barrier forms a second fluidic seal with the tubing string wall, where the second barrier is configured to open the second fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and where the UCA with the second barrier is configured to be inserted into the casing string after the LCA is inserted into the casing string.

[0007] In yet another aspect, the disclosure relates to a method of implementing a subsea completion system. The method can include inserting a lower completion assembly (LCA) into a casing string for a subsea wellbore, where the LCA includes a receptacle disposed toward its top end and has a LCA wall that forms a LCA cavity, where the LCA cavity has disposed therein toward its top end a first barrier, where first barrier forms a first substantial fluidic seal with the LCA wall, where the first barrier is configured to open the first substantial fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and where the LCA below the first barrier is filled with a working fluid. The method can also include filling the casing string with a packer fluid. The method can further include inserting an upper completion assembly (UCA) into the casing string, where the UCA includes a tubing string having a tubing string wall that forms a tubing string cavity, where the tubing string cavity has disposed therein toward its bottom end a second barrier, where the second barrier forms a second substantial fluidic seal with the tubing string wall, where the second barrier is configured to open the second substantial fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and where the UCA above the second barrier is filled with the working fluid. The method can also include engaging the bottom end of the UCA with the top end of the receptacle of the LCA. The method can further include activating a working fluid control system. The method can also include installing a crown plug.

[0008] These and other aspects, objects, features, and embodiments will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope, as the example embodiments may admit to other equally effective embodiments. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or positions may be exaggerated to help visually convey such principles. In the drawings, reference numerals designate like or corresponding, but not necessarily identical, elements.

[0010] FIG. **1** shows a field system that includes a subsea completion system according to certain example embodiments.

[0011] FIG. **2** shows details of the floating platform and the subsea completion system of the field system of FIG. **1** according to certain example embodiments.

[0012] FIGS. **3** through **6** show implementation of a subsea completion system according to certain example embodiments.

[0013] FIG. **7** shows a cross-sectional side view of a barrier used with an example subsea completion system according to certain example embodiments.

[0014] FIGS. **8** through **11** show implementation of another subsea completion system according to certain example embodiments.

[0015] FIG. **12** shows part of a subsea completion system according to certain example embodiments.
embodiments.

[0016] FIG. **13** shows a flowchart for a method of establishing a subsea completion system according to certain example embodiments.
[0017] FIG. **14** shows a flowchart for a method of implementing a subsea completion system according to certain example embodiments.

DETAILED DESCRIPTION

[0018] The example embodiments discussed herein are directed to systems, apparatus, methods, and devices for subsea completion systems. Example embodiments can be used in subsea completion systems for subterranean field operations (e.g., injection operations, production operations). Example embodiments can be used for subsea completion systems in offshore subterranean operations. While example embodiments are described as being used in conjunction with horizontal Xmas trees herein, example embodiments can be used in conjunction with other components and/or configurations of subsea completion systems.
[0019] Example subsea completion systems can include one or multiple components, where a component can be made from a single piece (as from a mold or an extrusion or a three-dimensional printing process). When a component (or portion thereof) of an example subsea completion system is made from a single piece, the single piece can be cut out, bent, stamped, and/or otherwise shaped to create certain features, elements, or other portions of the component. Alternatively, a component (or portion thereof) of an example subsea completion system can be made from multiple pieces that are mechanically coupled to each other. In such a case, the multiple pieces can be mechanically coupled to each other using one or more of a number of coupling methods, including but not limited to adhesives, welding, fastening devices, compression fittings, mating threads, and slotted fittings. One or more pieces that are mechanically coupled to each other can be coupled to each other in one or more of a number of ways, including but not limited to fixedly, hingedly, rotatably, removably, slidably, and threadably.
[0020] Example subsea completion systems can be designed to comply with certain standards and/or requirements. Examples of entities that set such standards and/or requirements can include, but are not limited to, the Society of Petroleum Engineers, the American Petroleum Institute (API), the International Standards Organization (ISO), and the Occupational Safety and Health Administration (OSHA). Each component of a subsea completion system (including portions thereof) can be made of one or more of a number of suitable materials, including but not limited to metal (e.g., stainless steel), ceramic, rubber, glass, fibrous material, and plastic.
[0021] The use of the terms “about”, “approximately”, and similar terms applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of ordinary skill in the art would consider as a reasonable amount of deviation to the recited numeric values (i.e., having the equivalent function or result). For example, this term may be construed as including a deviation of +10 percent of the given numeric value provided such a deviation does not alter the end function or result of the value. Therefore, a value of about 1% may be construed to be a range from 0.9% to 1.1%. Furthermore, a range may be construed to include the start and the end of the range. For example, a range of 10% to 20% (i.e., range of 10%-20%) includes 10% and also includes 20%, and includes percentages in between 10% and 20%, unless explicitly stated otherwise herein. Similarly, a range of between 10% and 20% (i.e., range between 10%-20%) includes 10% and also includes 20%, and includes percentages in between 10% and 20%, unless explicitly stated otherwise herein.
[0022] A “subterranean formation” refers to practically any volume under a surface. For example, it may be practically any volume under a terrestrial surface (e.g., a land surface), practically any volume under a seafloor, etc. Each subsurface volume of interest may have a variety of characteristics, such as petrophysical rock properties, reservoir fluid properties, reservoir conditions, hydrocarbon properties, or any combination thereof. For example, each subsurface volume of interest may be associated with one or more of: temperature, porosity, salinity, permeability, water composition, mineralogy, hydrocarbon type, hydrocarbon quantity, reservoir

location, pressure, etc. Those of ordinary skill in the art will appreciate that the characteristics are many, including, but not limited to: shale gas, shale oil, tight gas, tight oil, tight carbonate, carbonate, vuggy carbonate, sandstone, unconventional (e.g., a permeability of less than 25 millidarcy (mD) such as a permeability of from 0.000001 mD to 25 mD)), diatomite, geothermal, mineral, etc. The terms “formation”, “subsurface formation”, “hydrocarbon-bearing formation”, “reservoir”, “subsurface reservoir”, “subsurface area of interest”, “subsurface region of interest”, “subsurface volume of interest”, and the like may be used synonymously. The term “subterranean formation” is not limited to any description or configuration described herein.

[0023] A “well” or a “wellbore” refers to a single hole, usually cylindrical, that is drilled into a subsurface volume of interest. A well or a wellbore may be drilled in one or more directions. For example, a well or a wellbore may include a vertical well, a horizontal well, a deviated well, and/or other type of well. A well or a wellbore may be drilled in the subterranean formation for exploration and/or recovery of resources. A plurality of wells (e.g., tens to hundreds of wells) or a plurality of wellbores are often used in a field depending on the desired outcome.

[0024] A well or a wellbore may be drilled into a subsurface volume of interest using practically any drilling technique and equipment known in the art, such as geosteering, directional drilling, etc. Drilling the well may include using a tool, such as a drilling tool that includes a drill bit and a drill string. Drilling fluid, such as drilling mud, may be used while drilling in order to cool the drill tool and remove cuttings. Other tools may also be used while drilling or after drilling, such as measurement-while-drilling (MWD) tools, seismic-while-drilling tools, wireline tools, logging-while-drilling (LWD) tools, or other downhole tools. After drilling to a predetermined depth, the drill string and the drill bit may be removed, and then the casing, the tubing, and/or other equipment may be installed according to the design of the well. The equipment to be used in drilling the well may be dependent on the design of the well, the subterranean formation, the hydrocarbons, and/or other factors.

[0025] A well may include a plurality of components, such as, but not limited to, a casing, a liner, a tubing string, a sensor, a packer, a screen, a gravel pack, artificial lift equipment (e.g., an electric submersible pump (ESP)), and/or other components. If a well is drilled offshore, the well may include one or more of the previous components plus other offshore components, such as a riser. A well may also include equipment to control fluid flow into the well, control fluid flow out of the well, or any combination thereof. For example, a well may include a wellhead, a choke, a valve, and/or other control devices. These control devices may be located on the surface, in the subsurface (e.g., downhole in the well), or any combination thereof. In some embodiments, the same control devices may be used to control fluid flow into and out of the well.

[0026] In some embodiments, different devices (e.g., barriers as discussed below) may be used to control fluid flow into and out of a well. In some embodiments, the rate of flow of fluids through the well may depend on the fluid handling capacities of the surface facility that is in fluidic communication with the well. The equipment to be used in controlling fluid flow into and out of a well may be dependent on the well, the subsurface region, the surface facility, and/or other factors. Moreover, sand control equipment and/or sand monitoring equipment may also be installed (e.g., downhole and/or on the surface). A well may also include any completion hardware that is not discussed separately. The term “well” may be used synonymously with the terms “borehole,” “wellbore,” or “well bore.” The term “well” is not limited to any description or configuration described herein.

[0027] It is understood that when combinations, subsets, groups, etc. of elements are disclosed (e.g., combinations of components in a composition, or combinations of steps in a method), that while specific reference of each of the various individual and collective combinations and permutations of these elements may not be explicitly disclosed, each is specifically contemplated and described herein. By way of example, if an item is described herein as including a component of type A, a component of type B, a component of type C, or any combination thereof, it is understood that this phrase describes all of the various individual and collective combinations and permutations of these components. For example, in some embodiments, the item described by this phrase could include only a component of type A.

[0028] In some embodiments, the item described by this phrase could include only a component of type B. In some embodiments, the item described by this phrase could include only a component of type C. In some embodiments, the item described by this phrase could include a component of type A and a component of type B. In some embodiments, the item described by this phrase could include a component of type A and a component of type C. In some embodiments, the item described by this phrase could include a component of type B and a component of type C. In some embodiments, the item described by this phrase could include a component of type A, a component of type B, and a component of type C.

[0029] In some embodiments, the item described by this phrase could include two or more components of type A (e.g., A**1** and A**2**). In some embodiments, the item described by this phrase could include two or more components of type B (e.g., B**1** and B**2**). In some embodiments, the item described by this phrase could include two or more components of type C (e.g., C**1** and C**2**). In some embodiments, the item described by this phrase could include two or more of a first component (e.g., two or more components of type A (A**1** and A**2**)), optionally one or more of a second component (e.g., optionally one or more components of type B), and optionally one or more of a third component (e.g., optionally one or more components of type C).

[0030] In some embodiments, the item described by this phrase could include two or more of a first component (e.g., two or more components of type B (B**1** and B**2**)), optionally one or more of a second component (e.g., optionally one or more components of type A), and optionally one or more of a third component (e.g., optionally one or more components of type C). In some embodiments, the item described by this phrase could include two or more of a first component (e.g., two or more components of type C (C**1** and C**2**)), optionally one or more of a second component (e.g., optionally one or more components of type A), and optionally one or more of a third component (e.g., optionally one or more components of type B).

[0031] If a component of a figure is described but not expressly shown or labeled in that figure, the label used for a corresponding component in another figure can be inferred to that component. Conversely, if a component in a figure is labeled but not described, the description for such component can be substantially the same as the description for the corresponding component in another figure. The numbering scheme for the various components in the figures herein is such that each component is a three-digit number or a four-digit number, and corresponding components in other figures have the identical last two digits. For any figure shown and described herein, one or more of the components may be omitted, added, repeated, and/or substituted. Accordingly, embodiments shown in a particular figure should not be considered limited to the specific arrangements of components shown in such figure.

[0032] Further, a statement that a particular embodiment (e.g., as shown in a figure herein) does not have a particular feature or component does not mean, unless expressly stated, that such embodiment is not capable of having such feature or component. For example, for purposes of present or future claims herein, a feature or component that is described as not being included in an example embodiment shown in one or more particular drawings is capable of being included in one or more claims that correspond to such one or more particular drawings herein.

[0033] Example embodiments of subsea completion systems will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of subsea completion systems are shown. Subsea completion systems may, however, be embodied in many different forms and should not be construed as limited to the example embodiments set forth herein. Rather, these example embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of subsea completion systems to those of ordinary skill in the art. Like, but not necessarily the same, elements (also sometimes called components) in the various figures are denoted by like reference numerals for consistency.

[0034] Terms such as “first”, “second”, “outer”, “inner”, “top”, “bottom”, “above”, “below”, “distal”, “proximal”, “front,”, “rear,” “left,” “right,” “on”, and “within”, when present, are used merely to distinguish one component (or part of a component or state of a component) from another. This list of terms is not exclusive. Such terms are not meant to denote a preference or a particular orientation, and they are not meant to limit embodiments of subsea completion systems. In the following detailed description of the example embodiments, numerous specific details are set forth in order to provide a more thorough understanding of the invention. However, it will be apparent to one of ordinary skill in the art that the invention may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

[0035] FIG. **1** shows a field system **100** that includes a subsea completion system **140** according to certain example embodiments. FIG. **2** shows details of the floating structure **103** and the subsea completion system **140** of the field system **100** of FIG. **1** according to certain example embodiments. The field system **100** includes multiple components. For example, as shown in FIG. **1**, in addition to the subsea completion system **140** located in the water **194** at or near the seabed **102**, the field system **100** can include a wellbore **111** in a subterranean formation **110**, a floating structure **103** that floats in the water **194** at a waterline **193**, one or more controllers **104** located on the floating structure **103**, one or more sensor devices **160**, a working fluid control system **170** located on the floating structure **103**, and one or more users **175**, which can each include one or more user systems **176**, a driving system **180** located on the floating structure **103**, a subsea manifold **147** located at or near the seabed **102**, one or more pipelines **148** located at or near the seabed **102**, and other piping **188** located at or near the seabed **102**. The wellbore **111** is bounded around its outer perimeter by a casing string **115**.

[0036] The components shown in the field system **100** of FIGS. **1** and **2** are not exhaustive, and in some embodiments, one or more of the components shown in FIGS. **1** and **2** may not be included in the field system **100** in which example embodiments may be used. Any component of the field system **100** can be discrete or combined with one or more other components of the subsea completion system **140**. Also, one or more components of the field system **100** can have different configurations. Similarly, one or more of the components shown in FIGS. **1** and **2** may not be included in the subsea completion system **140** in which example embodiments may be used. Any component of the subsea completion system **140** can be discrete or combined with one or more other components of the subsea completion system **140**. Also, one or more components of the subsea completion system **140** can have different configurations.

[0037] The wellbore **111** has a casing string **115** (e.g., production casing), inside of which is positioned a tubing string. The tubing string has a cavity that extends continuously along its length, and there is an annulus between the tubing string and the casing string **115** that extends continuously along the length of the tubing string. The wellbore **111** may have any depth (e.g., thousands of feet, tens of thousands of feet). The diameter of the wellbore **111** (and so also the diameter of the casing string **115**) may decrease in stepwise increments along its length, starting with the largest diameter (e.g., 36 inches, 22 inches) at the seabed **102** and ending with the smallest diameter (e.g., 12 inches, 6 inches) at the distal end of the wellbore **111**. The wellbore **111** may have one or more vertical segments and/or one or more horizontal segments along its length. The true vertical depth (TVD) of the wellbore **111** may be less than the overall length of the wellbore **111**.

[0038] The floating structure **103** in this case is used for subterranean field operations (also called subsea field operations herein), in which exploration and production phases (also called stages) of the subsea field operation are executed to extract one or more subterranean resources (e.g., oil, natural gas, water, hydrogen gas) from and/or inject resources (e.g., carbon monoxide, carbon dioxide, water) into the subterranean formation **110** via the wellbore **111**. In some cases, as when the depth of the water **194** between the waterline **193** and the seabed **102** is relatively small (e.g., hundreds of feet), the structure **103** (e.g., in the form of a jack-up rig) may be mounted on the seabed **102**.

[0039] In alternative embodiments, as when a subsea operation is close to land, the structure **103** may be land-based rather than floating or otherwise in the water **194**. In addition, or in the alternative, a field operation involves multiple wellbores **111** that originate from the same proximate location (sometimes called a pad) on the seabed **102**. In such cases, the wellbores **111** are drilled and completed one at a time. Also, in such cases, there may be one subsea completion system **140** for each wellbore **111**. To extract a subterranean resource from a wellbore **111** on production and/or to inject a subterranean resource into the subterranean formation **110** through the wellbore **111**, the subsea completion system **140** is disposed toward the top of the wellbore **111** at the seabed **102**.

[0040] The subsea completion system **140** of the field system **100** may include or be an assembly (or multiple assemblies) of one or more of a number of components. For example, components of the subsea completion system **140** can include, but are not limited to, an upper wellhead, a tubing hanger **266**, a tubing spool, one or more valves, tubing pipes (part of a tubing string), a receptacle, one or more barriers **230** (discussed below), one or more valves, one or more sensor devices **160**, piping (e.g., piping **188**), a casing spool, and a casing hanger. Some of the components of the subsea completion system **140** may be grouped into their own assembly within the subsea completion system **140**.

[0041] For example, the subsea completion system **140** shown FIG. **2** includes an upper completion assembly **250**, a lower completion assembly **220**, and a subsea Xmas tree **177** located within part of the casing string **115**. The subsea Xmas tree **177** is a stack of vertical and/or horizontal valves, spools, pressure gauges, chokes, and/or other components installed as an assembly on a subsea wellhead. The subsea Xmas tree **177** may be configured to provide a controllable interface between the wellbore **111** and production facilities (e.g., via the subsea pipeline **148**). The various valves of the subsea Xmas tree **177** may be used for such purposes as testing, servicing, regulating, and/or choking the stream of produced subterranean resources coming up from the wellbore **111** and/or flowing down into the wellbore **111**. In certain example embodiments, the subsea Xmas tree **177** of the subsea completion system **140** is a horizontal tree. The subsea Xmas tree **177** may include a receiving feature that is configured to receive a tubing hanger **266**, which is integrated with the upper completion assembly **250**.

[0042] The lower completion assembly **220** (LCA **220**) of the subsea completion system **140** in this case includes a receptacle **225** disposed at a top end of the LCA **220**. The receptacle **225** has a receptacle wall **226** that forms a receptacle cavity **227** that runs along the height of the receptacle **225**. The receptacle **225** may take any of a number of forms. For example, the receptacle **225** may be or include a polished bore receptacle (PBR). The diameter of the receptacle wall **226** may be larger than the diameter (defined by a wall **224**) of the rest of the LCA **220** and larger than the diameter (defined by a wall **254**) of the tubing string **255** of the UCA **250**.

[0043] Depending on the location of the barrier **230-1**, the cavity **227** may continue downward beyond the receptacle **225** into part of the LCA **220** up to the barrier **230-1**. Below the barrier **230-1**, there is a continuous cavity **229** throughout the remainder of the LCA **220**. The remainder of the LCA **220** below the receptacle **225** may be or include a tubing string, which is made up of multiple tubing pipes that are coupled directly or indirectly to each other end to end in series (e.g., substantially similar to the tubing string **255** of the UCA **250**). Most of the tubing string of the LCA **220** may be positioned within the wellbore **111** in the subterranean formation **110**.

[0044] The LCA **220** of the subsea completion system **140** in this case also includes a barrier **230-1**. The barrier **230-1** of the LCA **220** is configured to be positioned within the cavity **229** of the LCA **220** and/or the cavity **227** of the receptacle **225**. The barrier **230-1** forms a fluidic seal with the adjacent wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**). In this way, the barrier **230-1** prevents the packer fluid **283** disposed between the LCA **220** and the casing string **115** from entering the LCA cavity **229** and, in some cases, some or all of the receptacle cavity **227**, when the barrier **230-1** forms a fluidic seal with the LCA wall **224** and/or the receptacle wall **226**. In this example, the barrier **230-1** also seals working fluid **282** within the LCA cavity **229** (which may include some of the receptacle cavity **227**) below the barrier **230-1** when the barrier **230-1** forms a fluidic seal with the LCA wall **224**.

[0045] The barrier **230-1** of the LCA **220** is not designed to be permanent. Specifically, the barrier **230-1** is designed to open (e.g., break) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) under certain conditions. In some cases, the barrier **230-1** may be configured in such a way that allows a user **175** to set the conditions under which the barrier **230-1** open (e.g., breaks) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**). In other cases, the conditions under which the barrier **230-1** opens (e.g., breaks) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) are set by the manufacturer of the barrier **230-1**.

[0046] As an example, the barrier **230-1** may be designed to open (e.g., break) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) when the differential force applied to the barrier **230-1** within the cavity **227** of the receptacle **225** and/or the cavity **229** of the remainder of the LCA **220** reaches a threshold value (e.g., 3000 psi, 6000 psi, 15000 psi). In other words, in such a configuration, the barrier **230-1** opens (e.g., breaks) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) when the difference in pressure within the cavity **227** and/or the cavity **229** on one side of the barrier **230-1** relative to the pressure within the cavity **227** and/or the cavity **229** on the other side of the barrier **230-1** exceeds the threshold value. In this case, the fluidic seal created between the barrier **230-1** and the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) isolates a working fluid **282** within the cavity **227** and/or the cavity **229** below the barrier **230-1** and a packer fluid **283** above the barrier **230-1** and outside the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**) within the casing string **115**.

[0047] The barrier **230-1** may be positioned at any point within the cavity **227** of the receptacle and/or the cavity **229** of the LCA **220**. In certain example embodiments, the barrier **230-1** is positioned within the cavity **227** and/or the cavity **229** toward the top end (e.g., within the receptacle **225**, just below the receptacle **225**) of the LCA **220**. In some cases, when the barrier **230-1** opens (e.g., breaks) the fluidic seal with the wall **224** of the LCA **220** (which may include the wall **226** of the receptacle **225**), the barrier **230-1** may break apart into a number of small pieces that are not retrievable. The barrier **230-1** may take one or more of a number of forms. For example, the barrier **230-1** may be or include a disc (e.g., made of glass, made of ceramics). As another example, the barrier **230-1** may be or include a valve (e.g., a multi-stage ball valve). An example of a barrier **230-1** is shown below with respect to FIG. **7**.

[0048] The upper completion assembly **250** (UCA **250**) of the subsea completion system **140** in this case includes a tubing string **255** that is or includes multiple tubing pipes that are coupled directly or indirectly to each other end to end in series. Depending on the location of the barrier **230-2**, some of the tubing string **255** may be disposed below the barrier **230-2** at the bottom end of the UCA **250**, while the remainder (the majority) of the tubing string **255** is disposed above the barrier **230-2**. The tubing string **255** has a tubing string wall **254** that forms a tubing string cavity **257** that runs along all or most of the height of the tubing string **255**. The diameter of the tubing string wall **254** may be smaller than the diameter formed by the receptacle wall **226** of the receptacle **225** of the LCA **220**. The cavity **257** originates at or near the floating structure **103** and continues downward to the barrier **230-2**. Most of the tubing string **255** of the UCA **250** is positioned inside the riser **116**. The UCA **250** also includes a tubing hanger **266** in this case.

[0049] The UCA **250** of the subsea completion system **140** in this case also includes a barrier **230-2**. The barrier **230-2** of the UCA **250** is configured to be positioned within the cavity **257** of the

UCA **250** and forms a fluidic seal with the adjacent wall **254** of the UCA **250**. In this way, the barrier **230-2** prevents the packer fluid **283** disposed between the UCA **250** and the casing string **115** from entering the tubing string cavity **257** when the barrier **230-2** forms a fluidic seal with the tubing string wall **254**. In this example, the barrier **230-2** also seals the working fluid **282** within the tubing string cavity **257** above the barrier **230-2** when the barrier **230-2** forms a fluidic seal with the tubing string wall **254**.

[0050] The barrier **230-2** of the UCA **250** is not designed to be permanent. Specifically, the barrier **230-2** is designed to open (e.g., break) the fluidic seal with the wall **254** of the UCA **250** under certain conditions. In some cases, the barrier **230-2** may be configured in such a way that allows a user **175** to set the conditions under which the barrier **230-2** opens (e.g., breaks) the fluidic seal with the wall **254** of the UCA **250**. In other cases, the conditions under which the barrier **230-2** opens (e.g., breaks) the fluidic seal with the wall **254** of the UCA **250** are set by the manufacturer of the barrier **230-2**.

[0051] As an example, the barrier **230-2** may be designed to open (e.g., break) the fluidic seal with the wall **254** of the UCA **250** when the differential force applied to the barrier **230-2** within the cavity **257** of the UCA **250** reaches a threshold value (e.g., 1500 psi, 3000 psi, 10000 psi). In other words, in such a configuration, the barrier **230-2** opens (e.g., breaks) the fluidic seal with the wall **254** of the UCA **250** when the difference in pressure within the cavity **257** on one side of the barrier **230-2** relative to the pressure within the cavity **257** on the other side of the barrier **230-2** exceeds the threshold value. In this case, the fluidic seal created between the barrier **230-2** and the wall **254** of the UCA **250** isolates a working fluid **282** within the cavity **257** above the barrier **230-2** and the packer fluid **283** below the barrier **230-2** and outside the wall **254** of the UCA **250** within the casing string **115**.

[0052] The barrier **230-2** may be positioned at any point within the cavity **257** of the UCA **250**. In certain example embodiments, the barrier **230-2** is positioned within the cavity **257** toward the bottom end (e.g., within the distal most tubing pipe of the tubing string **255**) of the UCA **250**. In some cases, when the barrier **230-2** opens (e.g., breaks) the fluidic seal with the wall **254** of the UCA **250**, the barrier **230-2** may break apart into a number of small pieces that are not retrievable. The barrier **230-2** may take one or more of a number of forms. For example, the barrier **230-2** may be or include a disc (e.g., made of glass, made of ceramics). As another example, the barrier **230-2** may be or include a valve (e.g., a multi-stage ball valve). As discussed above, an example of a barrier **230-2** is shown below with respect to FIG. **7**.

[0053] When the subsea completion system **140** includes the barrier **230-1** and the barrier **230-2**, at least one characteristic (e.g., threshold value, material, size, configuration, configurability) of the barrier **230-1** of the LCA **220** may differ from the corresponding characteristic of the barrier **230-2** of the UCA **250**. For example, the threshold value of the barrier **230-1** may be more than or less than the threshold value of the barrier **230-2**. As another example, the barrier **230-1** of the LCA **220** may be or include a ball valve, while the barrier **230-2** of the UCA **250** may be or include a glass disc.

[0054] Also positioned within the casing string **115** in FIG. **2** is a production packer **259**, which is generally a short hollow cylinder. The production packer **259** abuts against the inner surface of the wall of the casing string **115** along its outer perimeter, and the production packer **259** abuts against the outer surface of the LCA wall **224** along its inner perimeter. Under this configuration, the production packer **259** creates a fluidic seal with the wall **224** of the LCA **220** and with the casing string **115**. As a result, the packer fluid **283** remains above the production packer **259** within the casing string **115**, and none of the packer fluid **283** is below the production packer **259** within the casing string **115**.

[0055] Additional piping **188** may be used to transfer a subterranean resource between the subsea completion system **140** and a subsea manifold **147** that is located in the water **194**. The subsea manifold **147** is an arrangement of piping (e.g., piping **188**), valves, and/or other components that are configured to combine, distribute, control, monitor, and/or otherwise manipulate the fluid flowing through the piping **188** from the subsea completion system **140** and/or other subsea completion systems. The subsea manifold **147** may be a standalone component of the field system **100**. Alternatively, the subsea manifold **147** may be part of another component and/or subsystem (e.g., integrated with the subsea completion system **140**). In alternative embodiments, the field system **100** may include multiple subsea manifolds **147**, which may be arranged in series and/or in parallel with each other.

[0056] Additional piping **188** may be used to transfer a subterranean resource between the subsea manifold **147** and one or more subsea pipelines **148**. There may be one or more of a number of components and/or systems (e.g., a subsea electrical pump, a subsea compressor, a subsea process cooler) positioned between a subsea completion system **140** and the subsea pipelines **148** to assist in extracting a subterranean resource from the wellbore **111** and/or injecting a subterranean resource into the wellbore **111**. There may be one or more communication links **105** and/or power transfer links **187** between one or more of the subsea components (e.g., the subsea completion system **140**, a sensor device **160**, a subsea manifold **147**, one or more of the subsea pipelines **148**) and one or more components (e.g., a generator, a controller **104**, a compressor) disposed on the topsides of the floating structure **103** (or land-based structure **103**, as the case may be).

[0057] Each subsea pipeline **148** (also sometimes called a submarine pipeline **148**) is a series of pipes, coupled end to end, that is laid at or near to the seabed **102**. A subsea pipeline **148** moves a subterranean resource from the area of the wellbore **111** to some other location, typically for a midstream process (e.g., oil refining, natural gas processing). The piping **188**, also located subsea, may include multiple pipes, ducts, elbows, joints, sleeves, collars, and similar components that are coupled to each other (e.g., using coupling features such as mating threads) to establish a network for transporting the subterranean resource between the subsea completion system **140** and the subsea manifold **147**, and also between the subsea manifold **147** and one or more of the subsea pipelines **148**. While not shown in FIG. **1**, in alternative embodiments of the field system **100**, piping **188** may run from the floating structure **103** to one or more components (e.g., the subsea manifold **147**). Each component of the piping **188** may have an appropriate size (e.g., inner diameter, outer diameter) and be made of an appropriate material (e.g., steel) to safely and efficiently handle the pressure, temperature, flow rate, and other characteristics of the subterranean resource at the depth in the water **194**.

[0058] The riser **116** is positioned between the floating structure **103** and the subsea completion system **140**. The riser **116** isolates the water **194** from the fluids flowing between the floating structure **103** and the wellbore **111**. The riser **116** may be considered part of the casing string **115**. The riser **116** may have any length (e.g., hundreds of feet, thousands of feet) sufficient to span approximately between the waterline **193** and the seabed **102**. The riser **116** has a diameter sufficient to allow for fluid flow in either or both directions (e.g., toward the waterline **193**, toward the seabed **102**), which may include a tubing string.

[0059] A user **175** can be any person that interacts, directly or indirectly, with a controller **104**, a sensor device **160**, the driving system **180**, the working fluid control system **170**, a barrier **230**, and/or any other component of the field system **100**. Examples of a user **175** may include, but are not limited to, a company representative, an engineer, a geologist, a consultant, a contractor, and a manufacturer's representative. A user **175** can use one or more user systems **176**, which may include a display (e.g., a GUI). A user system **176** of a user **175** can interact with (e.g., send data to, obtain data from) a controller **104**, a sensor device **160**, the driving system **180**, and/or the working fluid control system **170** via an application interface and using the communication links **105**.

[0060] A user **175** can also interact directly with a controller **104**, a sensor device **160**, the driving system **180**, and/or the working fluid control system **170** through a user interface (e.g., keyboard, mouse, touchscreen). A user system **176** of a user **175** can interact with (e.g., sends data to, receives data from) a controller **104**, a sensor device **160**, the driving system **180**, and/or the working fluid control system **170** via an application interface. Examples of a user system **176** can include, but are not limited to, a cell phone with an app, a laptop computer, a handheld device, a smart watch, a desktop computer, and an electronic tablet.

[0061] The field system **100** can include one or more controllers **104**. A controller **104** of the system **100** communicates with and in some cases controls one or more of the other components (e.g., a sensor device **160**, the driving system **180**, the working fluid control system **170**) of the field system **100**. A controller **104** performs a number of functions that include obtaining and sending data, evaluating data, following protocols, running algorithms, and sending commands. A controller **104** can include one or more of a number of components. Such components of a controller **104** can include, but are not limited to, a control engine, a communication module, a timer, a counter, a power module, a storage repository, a hardware processor, memory, a transceiver, an application interface, and a security module. When there are multiple controllers **104** in the field system **100**, each controller **104** can operate independently of each other. Alternatively, one or more of the controllers **104** can work cooperatively with each other. As yet another alternative, one of the controllers **104** can control some or all of one or more other controllers **104** in the field system **100**. In some cases, a controller **104** is an optional component of the field system **100**.

[0062] Each sensor device **160** (e.g., optional sensor device **160-1**, optional sensor device **160-2**) includes one or more sensors that measure one or more parameters (e.g., pressure, distance, torque, flow rate, temperature, humidity, voltage, current). Examples of a sensor of a sensor device **160** can include, but are not limited to, a proximity sensor, a magnetic field sensor, a temperature sensor, torque sensor, a flow sensor, a pressure sensor, a gas spectrometer, a voltmeter, an ammeter, a permeability meter, a porosimeter, and a camera. A sensor device **160** can be integrated with and/or measure a parameter associated with one or more components (e.g., the UCA **250** of the subsea completion system **140**, the LCA **220** of the subsea completion system **140**) of the field system **100**. For example, a sensor device **160** can be configured to measure a parameter (e.g., torque, distance) associated with stabbing the distal end of the UCA **250** into the receptacle **225** at the top end of the LCA **220** of the subsea completion system **140**.

[0063] As another example, a sensor device **160** may be configured to provide an indication when the bottom end of the tubing string **255** of the UCA **250** is stabbed into the top end of the receptacle **225** of the LCA **220**. In such a case, the indication may trigger the working fluid control system **170** to raise the pressure through the working fluid **282** within the tubing string cavity **257** of the UCA **250** so that the differential force experienced by a barrier **230** (e.g., barrier **230-1**) exceeds a threshold value at which the barrier **230** opens (e.g., breaks apart). In some cases, the measurements made by one or more sensor devices **160**, each measuring a different parameter, can be used to determine and confirm whether a controller **104** should take a particular action (e.g., operate a valve, operate or adjust driving system **180**, operate or adjust the working fluid control system **170**). In some cases, a sensor device **160** can include its own controller (e.g., controller **104**), or portions thereof. In some cases, a sensor device **160** is an optional component of the field system **100**, or at least the subsea completion system **140**.

[0064] The driving system **180** may be configured to move (e.g., raise, lower, rotate) the UCA **250** within the casing string **115**, which may include the riser **116**. The driving system **180** may also be configured to insert the LCA **220** into position within the casing string **115** both above and below the seabed **102** prior to inserting and lowering the UCA **250** toward the LCA **220** within the casing string **115**. The driving system **180** may further be configured to lower (sometimes referred to as stabbing in) the bottom end of the tubing string **255** of the UCA **250** into the top end of the receptacle **225** of the LCA **220**.

[0065] The driving system **180** may include one or more of a number of components. Such components may include, but are not limited to, a motor, a gearbox, a compressor, a valve, a sensor device (e.g., sensor device **160**), piping (e.g., piping **188**), a controller (e.g., controller **104**), tongs, a rotating table, a drive shaft, and a protective relay. The driving system **180** may be controlled, in whole or in part, by one or more of the controllers **104**. In addition, or in the alternative, the driving system **180** may include its own controller that is configured to control some or all of the driving system **180**. In such a case, the controller of the driving system **180** may operate in coordination with one or more of the external controllers **104**.

[0066] The working fluid control system **170** of the field system **100** is configured to control the flow of the working fluid **282** into the UCA **250** from a top end of the UCA **250**. The working fluid control system **170** may also be configured to control (e.g., raise, lower) the pressure through the working fluid **282** within the tubing string cavity **257**. In this way, for example, the working fluid control system **170** may raise the pressure within the tubing string cavity **257** so that the differential pressure threshold for the barrier **230-1** is exceeded, thereby causing the barrier **230-1** to open (e.g., break apart), and so open the fluidic seal with the LCA wall **224** and/or the receptacle wall **226**. In such case, the barrier **230-2** may have previously been broken apart when a lower differential pressure, corresponding to the threshold value for the barrier **230-2**, had been exceeded.

[0067] The working fluid control system **170** may include one or more of a number of components. Such components may include, but are not limited to, a motor, a pump, a compressor, a valve, a sensor device (e.g., sensor device **160**), piping (e.g., piping **188**), a controller (e.g., controller **104**), a working fluid processing system, a working fluid storage system, a heater, a heat exchanger, and a protective relay. The working fluid control system **170** may be controlled, in whole or in part, by one or more of the controllers **104**. In addition, or in the alternative, the working fluid control system **170** may include its own controller that is configured to control some or all of the working fluid control system **170**. In such a case, the controller of the working fluid control system **170** may operate in coordination with one or more of the external controllers **104**.

[0068] Each communication link **105** can include wired (e.g., Class **1** electrical cables, electrical connectors, Power Line Carrier, RS485) and/or wireless (e.g., sound or pressure waves in the water **194**, Wi-Fi, Zigbee, visible light communication, cellular networking, Bluetooth, Bluetooth Low Energy (BLE), ultrawide band (UWB), WirelessHART, ISA100) technology. A communication link **105** can transmit signals (e.g., communication signals, control signals, data) from one component (e.g., a controller **104**) of the field system **100** to another (e.g., a valve on the Xmas tree **177**, the driving system **180**, the working fluid control system **170**).

[0069] Each power transfer link **187** can include one or more electrical conductors, which can be individual or part of one or more electrical cables. In some cases, as with inductive power, power can be transferred wirelessly using power transfer links **187**. A power transfer link **187** can transmit power from one component (e.g., a battery, a generator) of the field system **100** to another (e.g., a motor of the driving system **180**). Each power transfer link **187** can be sized (e.g., 12 gauge, 18 gauge, 4 gauge) in a manner suitable for the amount (e.g., 480V, 24V, 120V) and type (e.g., alternating current, direct current) of power transferred therethrough.

[0070] FIGS. **3** through **6** show implementation of a subsea completion system **340** over time according to certain example embodiments. Referring to the description above with respect to FIGS. **1** and **2**, the subsea completion system **340** shown in FIG. **3** includes a casing string **315** (which may include part of a riser, such as riser **116**), a production packer **359**, a LCA **320**, and a Xmas tree **377**. Each of these components (including portions thereof, such as the receptacle **325** of the LCA **320**) may be substantially the same as the corresponding component (or portion thereof) discussed above with respect to FIGS. **1** and **2**. The order of steps listed below with respect to FIGS. **3** through **6** may vary in different embodiments.

[0071] At the point in time captured in FIG. **3**, the LCA **320** has been inserted into the casing string **315**. The LCA **320** includes a receptacle **325** that is placed at the top end of the LCA **320**. The receptacle wall **326** has a diameter that is larger than the diameter of the LCA wall **324** and the diameter of the tubing string **355** of the UCA **450**. Positioned within the cavity **329** of the LCA **320** (and possibly within the cavity **327** of the receptacle **325**) is a barrier **330**, which creates a fluidic seal with the LCA wall **324** of the LCA **320** (which may include the receptacle wall **326** of the receptacle **325**). In this case, before the barrier **330** is placed within the cavity **329** of the LCA **320**, working fluid **382** is injected into the cavity **329** of the LCA **320**. The working fluid **382** may be injected into the cavity **329** of the LCA **320** using a working fluid control system (e.g., working fluid control system **170**).

[0072] After the LCA **320** is inserted into the casing string **315**, the production packer **359** may then be set within the casing string **315** toward the bottom of the casing string **315**. The production packer **359** may be a short hollow cylinder, as in this example. The production packer **359** abuts against the inner surface of the wall of the casing string **315** along its outer perimeter, and the production packer **359** abuts against the outer surface of the LCA wall **324** along its inner perimeter. Under this configuration, the production packer **359** creates a fluidic seal with the wall **324** of the LCA **320** and with the casing string **315**.

[0073] Prior to the production packer **359** being set, the Xmas tree **377** may be set in and around the casing string **315** above the LCA **320**. The Xmas tree **377** in this case is a horizontal tree with two valves set within the horizontal exit channel. The Xmas tree **377** in this example also has a shoulder onto which the tubing hanger **366** (discussed below) rests. The Xmas tree **377** may be put into place using various equipment, including but not limited to a crane, a remotely operated vehicle (ROV), and divers. Some or all of the Xmas tree **377** may be preassembled (e.g., on a production floor on land, on a floating structure **103**) prior to being submerged into the water **394**.

[0074] After the Xmas tree **377** is set, packer fluid **383** may be injected into the casing string **315**. The packer fluid **383** may be injected into the casing string **315** using a system that operates similarly to a working fluid control system **170**, but applied to packer fluid **383** rather than working fluid **382**. As a result of the fluidic seal created by the production packer **359**, the packer fluid **383** remains above the production packer **359** within the casing string **315**, and none of the packer fluid **383** is below the production packer **359** within the casing string **315**. The contents of the packer fluid **383** may be damaging to the subterranean formation (e.g., subterranean formation **110**), inhibiting the ability to extract subterranean resources. As a result, an objective of the field operation is to minimize the amount of packer fluid **383** that reaches the subterranean formation.

[0075] At the point of time captured in FIG. **4**, which follows the point in time captured in FIG. **3**, the UCA **450** may be inserted into the casing string **315**. The UCA **450** in this case includes a tubing string **355**, a tubing hanger **366**, and a barrier **430**. The barrier **430** is placed toward the distal (bottom) end of the UCA **450**. The position of the tubing hanger **366** on the UCA **450** is configured to coincide with the location of the Xmas tree **377** within the casing string **315** relative to the receptacle **325** and the barrier **330** of the LCA **320**. The tubing hanger **366** is positioned above the barrier **430** on the UCA **450**. The tubing hanger **366** includes 2 crown plugs **356**.

[0076] The cavity **357** of the tubing string **355** of the UCA **450** is filled with working fluid **382**. Since the barrier **430** creates a fluidic seal with the wall **354** of the tubing string **355**, the working fluid **382** remains above the barrier **430**, and none of the packer fluid **383** mixes with the working fluid **382** in the cavity **357**. The working fluid **382** may be injected into the cavity **357** of the tubing string **355** of the UCA **450** using a working fluid control system (e.g., working fluid control system **170**).

[0077] As shown in FIG. **4**, the distal end of the tubing string **355** of the UCA **450** has just been lowered into the top of the receptacle cavity **327** of the receptacle **325**. In certain example embodiments, as in this case, the inner surface of the receptacle wall **326** and/or the outer surface of the distal end of the tubing string **355** include one or more sealing members **391** (e.g., gaskets, O-rings) that create a fluidic seal, even as the tubing string **355** of the UCA **450** continues to be stabbed further downward within the casing string **315**. As a result, the cavity **327** of the receptacle **325**, filled with packer fluid **383** but sealed off by the sealing members **391** of the tubing string **355** and/or the receptacle **325**, becomes smaller and smaller as the UCA **450** is lowered further until the UCA **450** has reached its final position, which in this case is when the tubing hanger **366** is seated within the Xmas tree **377**. Further, the barrier **330** prevents any of the packer fluid **383** from mixing with the working fluid **382** within the cavity **329** of the LCA **320**, and the barrier **430** prevents any of the packer fluid **383** from mixing with the working fluid **382** within the cavity **357** of the tubing string **355** of the UCA **450**.

[0078] At the point of time captured in FIG. **5**, which follows the point in time captured in FIG. **4**, the UCA **450** is inserted even further into the casing string **315**. At this point in time, the UCA **450** is lowered further relative to what is shown in FIG. **4** until the tubing hanger **366** is seated within the Xmas tree **377**. At this point, the UCA **450** cannot be lowered any further. This position of the UCA **450** relative to the LCA **320** may additionally or alternatively be indicated by the use of one or more sensor devices (e.g., sensor device **160-1**, sensor service **160-2**). When this occurs, the size of the cavity **327** of the receptacle **325** is even smaller than what is shown in FIG. **4**. Since the cavity **327** is filled with packer fluid **383**, and since the sealing members **391** of the tubing string **355** and/or the receptacle **325** prevent the packer fluid **383** from escaping the shrinking cavity **327**, the resulting buildup of pressure within the cavity **327** applies a large differential force on the barrier **330** and the barrier **430**.

[0079] In this example, the threshold value of the differential force for the barrier **430** is lower than the threshold value of the differential force for the barrier **330**. Therefore, assuming the pressure in the cavity **357** of the tubing string **355** of the UCA **450** is substantially the same as the pressure in the cavity **329** of the LCA **320**, the barrier **330** and the barrier **430** experience substantially the same differential force. When the threshold value for the barrier **430** is lower than the threshold value for the barrier **330**, as in this case, the barrier **430** opens (e.g., breaks apart) when the differential force reaches the threshold value for the barrier **430** while the barrier **330** remains intact. When the barrier **430** opens by breaking apart, barrier remnants **621** are scattered in the cavity **327** above the barrier **330**. Also, the working fluid **382** in the cavity **357** of the tubing string **355** of the UCA **450** mixes with the packer fluid **383** in the cavity **327**. In certain example embodiments, the barrier **430** breaks apart into the barrier remnants **621** before the tubing hanger **366** lands on the Xmas tree **377**.

[0080] At the point of time captured in FIG. **6**, which follows the point in time captured in FIG. **5**, a sufficiently large differential force is applied to the barrier **330** through the working fluid **382** in the cavity **357** of the tubing string **355** of the UCA **450** to open (e.g., break) the barrier **330**, resulting in barrier remnants **721** of the barrier **330** to remain in the now combined cavity that includes cavity **357**, cavity **327**, and cavity **329**. The working fluid **382** that was previously in the cavity **357** in the UCA **450** and the working fluid **382** that was previously in the cavity **329** of the LCA **320** are now combined, and the combined cavity spans the entire height of the casing string **315**.

[0081] Using this example process shown and described with respect to FIGS. **3** through **6**, the crown plugs **356**, pre-installed (e.g., on land), are run quickly, safely, and effectively, saving 3 to 5 days of installing them using methods currently known in the art. This example process also saves tremendous amounts of money and avoids tremendous safety risks that apply to personnel and equipment. For instance, this example process avoids the need to rig up a CCTLF, rig up a SFT tool, rig up a slickline, run the slickline to retrieve the isolation sleeve, install the crown plugs, rig down the slickline, rig down the SFT, and rig down the CCTLF.

[0082] FIG. **7** shows a cross-sectional side view of an example of a barrier **730** used with an example subsea completion system according to certain example embodiments. Referring to the description above with respect to FIGS. **1** through **6**, the barrier **730** of FIG. **7** is an example of the barriers (e.g., barrier **330**) discussed above. In this case, the barrier **730** is a sub that is configured to be integrated with a tubing string (e.g., tubing string **355**) or a LCA (e.g., LCA **320**). For example, the barrier **730** in the form of a sub may have mating threads that allow the barrier **730** to be threadably coupled to a tubing pipe on both ends so that the barrier **730** is in line with the adjacent tubing pipes as part of the tubing string.

[0083] The barrier **730** may be a specially designed product or an off-the-shelf product that is available in the current art. In this case, however, the barrier **730** may be inverted from its normal orientation during use in the current art. Specifically, the upper housing **737** (also called the box) in this case may be positioned at the downstream end, and the lower housing **738** (also called the pin) in this case may be positioned at the upstream end. The barrier **730** has a wall **731** that forms a cavity **739** along its length. Inside the cavity **739** are positioned a disc **735** (e.g., made of glass, made of ceramics), one or more shear pins **734**, a piercing device **732**, and a sliding sleeve **733**.

[0084] Each shear pin **734** is configured to shear when a differential force applied to the disc **735** within the cavity **739** (which coincides with the cavity (e.g., cavity **329**, cavity **357**) of the LCA (e.g., LCA **320**) or UCA (e.g., UCA **450**) into which the barrier **730** is integrated) reaches a threshold value. Each shear pin **734** has one end anchored in an aperture in the inner surface of the wall **731** of the barrier **730**. The other end of the shear pin **734** is positioned in an aperture in the wall of the sliding sleeve **733**. When at least one shear pin **734** is intact, the shear pin **734** fixes the position of the sliding sleeve **733** within the cavity **739**.

[0085] The sliding sleeve **733** does not move until all shear pins **734** have sheared. As more shear pins **734** are used with the barrier **730**, the threshold value of the differential force required to shear the shear pins **734** increases. In this way, the threshold value of the differential force to activate (e.g., open, break, shear) the barrier **730** may be calibrated (e.g., by a user in the field, at a manufacturer's facility) or otherwise tailored for the operating conditions of a particular field system (e.g., field system **100**).

[0086] The sliding sleeve **733** of the barrier **730** has a limited range of motion within an elongated slot or recess in the inner surface of the wall **731**. As discussed above, if at least one shear pin **734** is intact, the position of the sliding sleeve **733** remains fixed within the cavity **739**. If all of the shear pins **734** have sheared, then the sliding sleeve **733** is free to move within the cavity **739**, albeit within a limited range of motion. When the orientation of the barrier **730** is inverted, as in this case, pressure building up in the cavity **739** as the UCA (e.g., UCA **450**) is lowered causes the disc **735** to apply an upward force to the sliding sleeve **733**. When the upward force applied by the disc **735** overcomes the shear pins **734**, the glass disc **735** and sliding sleeve **733** move upwards together. In its upward travel, when the disc **735** contacts the piercing device **732**, the disc **735** breaks, which opens (e.g., breaks) the fluidic seal that the disc **735** maintained to that point in time.

[0087] The piercing device **732** is used to break the disc **735** when the barrier **730** is installed in a normal position (upside down from what is shown in FIG. **7**). Specifically, in a normal orientation of the barrier **730**, when all of the shear pins **734** shear, the sliding sleeve **733** moves away from the disc **735**, allowing the disc **735** to follow the sliding sleeve **733**. The piercing device **732** is fixed in the wall **731** and sticks out enough from the wall **731** to be contacted by the disc **735** as the disc **735** follows the sliding sleeve **733**. When the piercing device **732** is contacted by the disc **735**, the disc **735** breaks.

[0088] The disc **735** may be made of any of a number of suitable materials (e.g., glass, ceramics). The thickness of the disc **735** is large enough to withstand the differential forces required to maintain a fluidic seal up until the threshold value set for the barrier **730**. When the barrier **730** is oriented in an inverted position, the disc **735** may be configured to break when the disc **735**, after applying a sufficiently strong upward force against the sliding sleeve **733** to overcome the shear pins **734**, collides with the piercing device **732**.

[0089] FIGS. **8** through **11** show implementation of another subsea completion system **840** according to certain example embodiments. Referring to the description above with respect to FIGS. **1** through **7**, the subsea completion system **840** shown in FIG. **8** includes a casing string **815** (which may include part of a riser, such as riser **116**), a production packer **859**, a LCA **820**, a UCA **850**, and a Xmas tree **877**. Each of these components (including portions thereof, such as the receptacle **825** of the LCA **820** or the tubing hanger **866** and crown plugs **856** of the UCA **850**) may be substantially the same as the corresponding component (or portion thereof) discussed above with respect to FIGS. **1** through **7**. The order of steps listed below with respect to FIGS. **8** through

11 may vary in different embodiments.

[0090] At the point in time captured in FIG. **8**, the LCA **820** has been inserted into the casing string **815**. The LCA **820** includes a receptacle **825** that is placed at the top end of the LCA **820**. The receptacle wall **826** has a diameter that is larger than the diameter of the LCA wall **824** and the diameter of the tubing string **855** of the UCA **850**. Positioned within the cavity **829** of the LCA **820** (and possibly within the cavity **827** of the receptacle **825**) is a barrier **830-1**, which creates a fluidic seal with the LCA wall **824** of the LCA **820** (which may include the receptacle wall **826** of the receptacle **825**). In this case, before the barrier **830-1** is placed within the cavity **829** of the LCA **820**, working fluid **882** is injected into the cavity **829** of the LCA **820**. The working fluid **882** may be injected into the cavity **829** of the LCA **820** using a working fluid control system (e.g., working fluid control system **170**).

[0091] After the LCA **820** is inserted into the casing string **815**, the production packer **859** may then be set within the casing string **815** toward the bottom of the casing string **815**. The production packer **859** may be a short hollow cylinder, as in this example. The production packer **859** abuts against the inner surface of the wall of the casing string **815** along its outer perimeter, and the production packer **859** abuts against the outer surface of the LCA wall **824** along its inner perimeter. Under this configuration, the production packer **859** creates a fluidic seal with the wall **824** of the LCA **820** and with the casing string **815**.

[0092] After the production packer **859** is set, the Xmas tree **877** may be set in and around the casing string **815** above the LCA **820**. The Xmas tree **877** in this case may be a horizontal tree. The Xmas tree **877** in this example also has a shoulder onto which the tubing hanger **866** (discussed below) rests. The Xmas tree **877** may be put into place using various equipment, including but not limited to a crane, a remotely operated vehicle (ROV), and divers. Some or all of the Xmas tree **877** may be preassembled (e.g., on a production floor on land, on a floating structure **103**) prior to being submerged into the water **894**.

[0093] After the Xmas tree **877** is set, packer fluid **883** may be injected into the casing string **815**. The packer fluid **883** may be injected into the casing string **815** using a system that operates similarly to a working fluid control system **170**, but applied to packer fluid **883** rather than the working fluid **882**. As a result of the fluidic seal created by the production packer **859** and the barrier **830-1**, the packer fluid **883** remains above the production packer **859** within the casing string **815**, and none of the packer fluid **883** is below the production packer **859** within the casing string **815**. The contents of the packer fluid **883** may be damaging to the subterranean formation (e.g., subterranean formation **110**), inhibiting the ability to extract subterranean resources. As a result, an objective of the field operation is to minimize the amount of packer fluid **883** that reaches the subterranean formation.

[0094] After the packer fluid **883** is injected into the casing string **815**, the UCA **850** may be inserted into the casing string **815**. The UCA **850** in this case includes a tubing string **855**, a tubing hanger **866**, and a barrier **830-2**. The barrier **830-2** is placed toward the distal (bottom) end of the UCA **850**. The position of the tubing hanger **866** on the UCA **850** is configured to coincide with the location of the Xmas tree **877** within the casing string **815** relative to the receptacle **825** and the barrier **830-1** of the LCA **820**. The tubing hanger **866** is positioned above the barrier **830-2** on the UCA **850**. The tubing hanger **866** includes 2 crown plugs **856**.

[0095] The cavity **857** of the tubing string **855** of the UCA **850** is filled with working fluid **882**. Since the barrier **830-2** creates a fluidic seal with the wall **854** of the casing string **855**, the working fluid **882** remains above the barrier **830-2**, and none of the packer fluid **883** mixes with the working fluid **882** in the cavity **857**. The working fluid **882** may be injected into the cavity **857** of the tubing string **855** of the UCA **850** using a working fluid control system (e.g., working fluid control system **170**).

[0096] As shown in FIG. **9**, the distal end of the tubing string **855** of the UCA **850** has just been lowered into the top of the receptacle cavity **827** of the receptacle **825**. In certain example embodiments, as in this case, the inner surface of the receptacle wall **826** and/or the outer surface of the distal end of the tubing string **855** include one or more sealing members **891** (e.g., gaskets, O-rings) that create a fluidic seal, even as the tubing string **855** of the UCA **850** continues to be stabbed further downward within the casing string **815**. As a result, the cavity **827** of the receptacle **825**, filled with packer fluid **883** but sealed off by the sealing members **891** of the tubing string **855** and/or the receptacle **825**, becomes smaller and smaller as the UCA **850** is lowered further until the UCA **850** has reached its final position, which in this case is when the tubing hanger **866** is seated within the Xmas tree **877**. Further, the barrier **830-1** prevents any of the packer fluid **883** from mixing with the working fluid **882** within the cavity **829** of the LCA **820**, and the barrier **830-2** prevents any of the packer fluid **883** from mixing with the working fluid **882** within the cavity **857** of the tubing string **855** of the UCA **850**.

[0097] At the point of time captured in FIG. **10**, which follows the point in time captured in FIG. **9**, the UCA **850** is inserted even further into the casing string **815**. At this point in time, the UCA **850** is lowered further relative to what is shown in FIG. **8** until the tubing hanger **866** is seated within the Xmas tree **877**. At this point, the UCA **850** cannot be lowered any further. This position of the UCA **850** relative to the LCA **820** may additionally or alternatively be indicated by the use of one or more sensor devices (e.g., sensor device **160-1**, sensor device **160-2**).

[0098] Just prior to the time shown in FIG. **10**, and just after the time shown in FIG. **9**, the distal end of the UCA **850** contacts and opens (e.g., breaks) the barrier **830-1**. In other words, the momentum of the distal end of the UCA **850** provides sufficient differential force to the barrier **830-1** to exceed the threshold value for the barrier **830-1**. As a result, as shown in FIG. **10**, the barrier **830-1** is gone and barrier remnants **1021** of the barrier **830-1** are shown in the cavity **829**. The barrier remnants **1021** are located below the barrier **830-2**. Also, the packer fluid **883** in the cavity **827** of the receptacle **825** mixes with the packer fluid **882** in the cavity **829**.

[0099] In this example, the threshold value of the differential force for the barrier **830-1** may be lower or higher than the threshold value of the differential force for the barrier **830-2**. At the time shown in FIG. **10**, the barrier **830-2** of the UCA **850** remains intact. Consequently, the differential force applied to the barrier **830-2** as the UCA **850** is stabbed into the receptacle **825** of the LCA **820** and as the distal end of the UCA **850** opens (e.g., breaks) the barrier **830-1** is not sufficient to exceed the threshold value of the barrier **830-2**.

[0100] At the point of time captured in FIG. **11**, which follows the point in time captured in FIG. **10**, a sufficiently large differential force is applied to the barrier **830-2** through the working fluid **882** in the cavity **857** of the tubing string **855** of the UCA **850** to open (in this case, break) the barrier **830-2**, resulting in barrier remnants **1121** of the barrier **830-2** to remain in the now combined cavity that includes cavity **857**, cavity **827**, and cavity **829**. The working fluid **882** that was previously in the cavity **857** in the UCA **850** and the working fluid **882** that was previously in the cavity **829** of the LCA **820** are now combined, and the combined cavity spans the entire height of the casing string **815**.

[0101] Using this example process shown and described with respect to FIGS. **8** through **11**, the crown plugs **856** are run and installed quickly, safely, and effectively, saving 3 to 5 days of down time installing them using methods currently known in the art. This example process also saves tremendous amounts of money and avoids tremendous safety risks that apply to personnel and equipment. For instance, this example process avoids the need to rig up a CCTLF, rig up a SFT tool, rig up a slickline, run the slickline to retrieve the isolation sleeve, install the crown plugs, rig down the slickline, rig down the SFT, and rig down the CCTLF.

[0102] FIG. **12** shows part of a subsea completion system **1240** according to certain example embodiments. Referring to the description above with respect to FIGS. **1** through **11**, the subsea completion system **1240** shown in FIG. **12** is substantially similar to the subsea completion system **140** discussed above with respect to FIG. **2**, except as described below. For example, the subsea completion system **1240** shown in FIG. **12** includes a UCA **1250**, a LCA **1220**, and a subsea Xmas tree **1277** located within part of the casing string **1215**, which itself is located, at least in part, in water **1294**. The subsea Xmas tree **1277** of the subsea completion system **1240** in this case may be a horizontal tree.

[0103] The LCA **1220** of the subsea completion system **1240** in this case includes a receptacle **1225** disposed at a top end of the LCA **1220**. The receptacle **1225** has a receptacle wall **1226** that forms a receptacle cavity **1227** that runs along the height of the receptacle **1225**. The receptacle **1225** may take any of a number of forms. For example, the receptacle **225** may be or include a polished bore receptacle (PBR). The diameter of the receptacle wall **1226** may be larger than the diameter (defined by a wall **1224**) of the rest of the LCA **1220** and larger than the diameter (defined by a wall **1254**) of the tubing string **1255** of the UCA **1250**.

[0104] In this example, there may be multiple barriers **1230** integrated with the LCA **1220**. Specifically, there are X barriers **1230** (barrier **1230-1** through barrier **1230-**X) within the receptacle cavity **1227** of the receptacle **1225** and/or the cavity **1229** of the remainder of the LCA **1220**. The barriers **1230** are aligned in series with each other. Adjacent barriers **1230** are separated from each other by a distance within the receptacle cavity **1227** of the receptacle **1225** and/or the cavity **1229** of the remainder of the LCA **1220**. The characteristics (e.g., thickness, material, configurability, configuration, differential force threshold value) of one barrier **1230** may be the same as, or different than, the characteristics of one or more of the other barriers **1230**. Each barrier **1230** forms a fluidic seal with the adjacent wall **1224** of the LCA **1220** (which may include the wall **1226** of the receptacle **1225**).

[0105] Below the lowest barrier **1230-**X, there is a continuous cavity **1229** throughout the remainder of the LCA **1220**. The remainder of the LCA **1220** below the receptacle **1225** may be or include a tubing string, which is made up of multiple tubing pipes that are coupled directly or indirectly to each other end to end in series (e.g., substantially similar to the tubing string **1255** of the UCA **1250**). Most of the tubing string of the LCA **1220** may be positioned within the wellbore (e.g., wellbore **111**) in the subterranean formation (e.g., subterranean formation **110**). In this case, the cavity **1229** above the lowest barrier **1230-**X is filled with working fluid **1282**. There is no working fluid **1282** above the highest barrier **1230-1**. There may be, or may not be, working fluid **1282** between intermediate barriers **1230** disposed in the cavity **1229**.

[0106] Each barrier **1230** of the LCA **1220** is not designed to be permanent. Specifically, each barrier **1230** is designed to open (e.g., break) the fluidic seal with the wall **1224** of the LCA **1220** (which may include the wall **1226** of the receptacle **1225**) under certain conditions. In some cases, a barrier **1230** may be configured in such a way that allows a user **175** to set the conditions under which the barrier **1230** opens (e.g., breaks) the fluidic seal with the wall **1224** of the LCA **1220** (which may include the wall **1226** of the receptacle **1225**). In other cases, the conditions under which a barrier **1230** opens (e.g., breaks) the fluidic seal with the wall **1224** of the LCA **1220** (which may include the wall **1226** of the receptacle **1225**) are set by the manufacturer of the barrier **1230**. In certain example embodiments, the threshold value of the differential force of a barrier **1230** is less than the threshold value of the differential force of the next-lowest barrier **1230** (if any) and is greater than the threshold value of the differential force of the next-highest barrier **1230** (if any).

[0107] Each barrier **1230** may be positioned at any point within the cavity **1227** of the receptacle **1225** and/or the cavity **1229** of the LCA **1220**. In some cases, when a barrier **1230** opens (e.g., breaks) the fluidic seal with the wall **1224** of the LCA **1220** (which may include the wall **1226** of the receptacle **1225**), the barrier **1230** may break apart into a number of small pieces that are not retrievable. A barrier **1230** may take one or more of a number of forms. For example, a barrier **1230** may be or include a disc (e.g., made of glass, made of ceramics). As another example, a barrier **1230** may be or include a valve (e.g., a multi-stage ball valve).

[0108] The UCA **1250** of the subsea completion system **1240** in this case includes a tubing string **1255** that is or includes multiple tubing pipes that are coupled directly or indirectly to each other end to end in series. Some of the tubing string **1255** may be disposed below the lowest-most barrier **1330-1** at the bottom end of the UCA **1250**, while the remainder (the majority) of the tubing string **1255** is disposed above the lowest-most barrier **1330-1**. The tubing string **1255** has a tubing string wall **1254** that forms a tubing string cavity **1257** that runs along all or most of the height of the tubing string **1255**. The diameter of the tubing string wall **1254** may be smaller than the diameter formed by the receptacle wall **1226** of the receptacle **1225** of the LCA **1220**. The cavity **1257** may originate at or near a floating platform (e.g., floating structure **103**) and continues downward to the upper-most barrier **1330**-Y. Most of the tubing string **1255** of the UCA **1250** may be positioned inside a riser (e.g., riser **116**). The UCA **1250** also includes a tubing hanger **1266**.

[0109] In this example, there may be multiple barriers **1330** integrated with the UCA **1250**. Specifically, there are Y barriers **1330** (barrier **1330-1** through barrier **1330**-Y) within the cavity **1257** of the tubing string **1255** of the UCA **1250**. The barriers **1330** are aligned in series with each other. Adjacent barriers **1330** are separated from each other by a distance within the cavity **1257** of the tubing string **1255** of the UCA **1250**. The characteristics (e.g., thickness, material, configurability, configuration, differential force threshold value) of one barrier **1330** may be the same as, or different than, the characteristics of one or more of the other barriers **1330**. Each barrier **1330** forms a fluidic seal with the adjacent wall **1254** of the UCA **1250**. Above the highest barrier **1330**-Y, there is a continuous cavity **1257** throughout the remainder of the tubing string **1255** (which may also include a riser (e.g., riser **116**)) of the UCA **1250**. In this case, the cavity **1257** above the highest barrier **1330**-Y is filled with working fluid **1282**. There is no working fluid **1282** below the lowest barrier **1330-1**. There may be, or may not be, working fluid **1282** between intermediate barriers **1330** disposed in the cavity **1257**.

[0110] Each barrier **1330** of the UCA **1250** is not designed to be permanent. Specifically, each barrier **1330** is designed to open (e.g., break) the fluidic seal with the wall **1254** of the UCA **1250** under certain conditions. In some cases, a barrier **1330** may be configured in such a way that allows a user (e.g., user **175**) to set the conditions under which the barrier **1330** opens (e.g., breaks) the fluidic seal with the wall **1254** of the UCA **1250**. In other cases, the conditions under which a barrier **1330** opens (e.g., breaks) the fluidic seal with the wall **1254** of the UCA **1250** are set by the manufacturer of the barrier **1330**. Each barrier **1330** may be positioned at any point within the cavity **1257** of the UCA **1250**. In some cases, when a barrier **1330** opens (e.g., breaks) the fluidic seal with the wall **1254** of the UCA **1250**, the barrier **1330** may break apart into a number of small pieces that are not retrievable. A barrier **1330** may take one or more of a number of forms. For example, a barrier **1330** may be or include a disc (e.g., made of glass, made of ceramics). As another example, a barrier **1330** may be or include a valve (e.g., a multi-stage ball valve).

[0111] Also positioned within the casing string **1215** in FIG. **12** is a production packer **1259**, which is generally a short hollow cylinder. The production packer **1259** abuts against the inner surface of the wall of the casing string **1215** along its outer perimeter, and the production packer **1259** abuts against the outer surface of the LCA wall **1224** along its inner perimeter. Under this configuration, the production packer **1259** creates a fluidic seal with the wall **1224** of the LCA **1220** and with the casing string **1215**. As a result, the packer fluid **1283** remains above the production packer **1259** within the casing string **1215**, and none of the packer fluid **1283** is below the production packer **1259** within the casing string **1215**.

[0112] FIG. **13** shows a flowchart **1397** for a method of establishing a subsea completion system according to certain example embodiments. While the various steps in this flowchart **1397** are presented sequentially, one of ordinary skill will appreciate that some or all of the steps may be executed in different orders, may be combined or omitted, and some or all of the steps may be executed in parallel. Further, in one or more of the example embodiments, one or more of the steps shown in this example method may be omitted, repeated, and/or performed in a different order. Some or all of the steps of the method of FIG. **13** can be performed off site (e.g., at a manufacturing facility, at a staging area). In addition, or in the alternative, some or all of the steps of the method of FIG. **13** can be performed on site (e.g., in the field, on a floating structure **103**) where a field operation is being performed or planned.

[0113] In addition, a person of ordinary skill in the art will appreciate that additional steps not shown in FIG. **13** may be included in performing this method. Accordingly, the specific arrangement of steps should not be construed as limiting the scope. Further, a particular computing device, such as a controller **104** discussed above with respect to FIG. **1**, can be used to perform one or more of the steps (or portions thereof) for the method shown in FIG. **13** in certain example embodiments. Any of the functions performed below by a controller **104** can involve the use of one or more protocols, one or more algorithms, and/or stored data stored in a storage repository. In addition, or in the alternative, any of the functions in the method can be performed by a user (e.g., user **175**), which may include an associated user system (e.g., user system **176**).

[0114] The method shown in FIG. **13** is merely an example that can be performed by using an example system described herein. In other words, systems for establishing a subsea completion system **140** can perform other functions using other methods in addition to and/or aside from those shown in FIG. **13**. Referring to the description above with respect to FIGS. **1** through **12**, but using reference numbers from FIGS. **1** and **2** by way of example in this case, the method shown in the flowchart **1397** of FIG. **13** begins at the START step and proceeds to step **1361**, where a first barrier **230-1** is calibrated. The first barrier **230-1** may be calibrated to open (e.g., break) at approximately a threshold value. The threshold value may correspond to an amount of a differential force applied to the first barrier **230-1**. When the first barrier **230-1** opens (e.g., breaks), a fluidic seal formed by the first barrier **230-1** is opened (e.g., broken). The first barrier **230-1** may be calibrated by a user **175** (e.g., in the field, at an offsite facility). Calibrating the first barrier **230-1** may involve whatever actions (e.g., insertion of retention pins, rotating a dial) are required based on the configuration of the first barrier **230-1**. In some cases, the first barrier **230-1** is calibrated at the factory, and this calibration setting may not subsequently be changed. In such cases, this step **1361** may be optional.

[0115] In step **1362**, the first barrier **230-1** is inserted within a top end of a lower completion assembly (LCA) cavity **229** formed by a LCA wall **224** of the LCA **220**. The first barrier **230-1** may form a fluidic seal with the LCA wall **224**. The first barrier **230-1** may be configured (e.g., based on construction, based on a factory setting, based on the calibration in step **1361**) to open (e.g., break) when the first barrier **230-1** has a differential force with a threshold value applied to it within the LCA cavity **229**. The LCA **220** with the first barrier **230-1** may be configured to be subsequently inserted into the casing string **115**. In some cases, as when the first barrier **230-1** is a sub as shown and described in FIG. **7** above (e.g., including a glass disc **735**, a sliding sleeve **733**, one or more shear pins **734** used to hold the sliding sleeve **733** in place), inserting the first barrier **230-1** may include inverting a standard orientation of the sub. Alternatively, the first barrier **230-1** may be installed in a standard orientation.

[0116] In step **1363**, a second barrier **230-2** is calibrated. The second barrier **230-2** may be calibrated to open (e.g., break) at approximately a threshold value. The threshold value may correspond to an amount of a differential force applied to the second barrier **230-2**. When the second barrier **230-2** opens (e.g., breaks), a fluidic seal formed by the second barrier **230-2** is opened (broken). The characteristics (e.g., thickness, material, configurability, configuration, differential force threshold value) of the second barrier **230-2** may be the same as, or different than, the characteristics of the first barrier **230-1**.

[0117] The second barrier **230-2** may be calibrated by a user **175** (e.g., in the field, at an offsite facility). Calibrating the second barrier **230-2** may involve whatever actions (e.g., insertion of retention pins, rotating a dial) are required based on the configuration of the second barrier **230-2**. The threshold value to which the second barrier **230-2** is calibrated may be different than the threshold value to which the first barrier **230-1** is calibrated in certain example embodiments. In some cases, the second barrier **230-2** is calibrated at the factory, and this calibration setting may not subsequently be changed. In such cases, this step **1363** may be optional.

[0118] In step **1364**, the second barrier **230-2** is inserted within a bottom end of an upper completion assembly (UCA) cavity **257** formed by a tubing string wall **254** of a tubing string **255** of the UCA **250**. The second barrier **230-2** may form a fluidic seal with the tubing string wall **254** of the UCA **250**. The second barrier **230-2** may be configured (e.g., based on construction, based on a factory setting, based on the calibration in step **1363**) to open (e.g., break) when the second barrier **230-2** has a differential force with a threshold value applied to it within the UCA cavity **257**. The UCA **250** with the second barrier **230-2** may be configured to be subsequently inserted into the casing string **115**.

[0119] In some cases, as when the second barrier **230-2** is a sub as shown and described in FIG. **7** above (e.g., including a glass disc **735**, a sliding sleeve **733**, one or more shear pins **734** used to hold the sliding sleeve **733** in place), inserting the second barrier **230-2** may include inverting a standard orientation of the sub. Alternatively, the second barrier **230-2** may be installed in a standard orientation. When step **1364** is complete, the subsea completion system **140** is established, and the process proceeds to the END step.

[0120] FIG. **14** shows a flowchart **1497** for a method of implementing a subsea completion system according to certain example embodiments. While the various steps in this flowchart **1497** are presented sequentially, one of ordinary skill will appreciate that some or all of the steps may be executed in different orders, may be combined or omitted, and some or all of the steps may be executed in parallel. Further, in one or more of the example embodiments, one or more of the steps shown in this example method may be omitted, repeated, and/or performed in a different order. Some or all of the steps of the method of FIG. **14** can be performed off site (e.g., at a manufacturing facility, at a staging area). In addition, or in the alternative, some or all of the steps of the method of FIG. **14** can be performed on site (e.g., in the field, on a floating structure **103**) where a field operation is being performed or planned.

[0121] In addition, a person of ordinary skill in the art will appreciate that additional steps not shown in FIG. **14** may be included in performing this method. Accordingly, the specific arrangement of steps should not be construed as limiting the scope. Further, a particular computing device, such as a controller **104** discussed above with respect to FIG. **1**, can be used to perform one or more of the steps (or portions thereof) for the method shown in FIG. **14** in certain example embodiments. Any of the functions performed below by a controller **104** can involve the use of one or more protocols, one or more algorithms, and/or stored data stored in a storage repository. In addition, or in the alternative, any of the functions in the method can be performed by a user (e.g., user **175**), which may include an associated user system (e.g., user system **176**).

[0122] The method shown in FIG. **14** is merely an example that can be performed by using an example system described herein. In other words, systems for implementing a subsea completion system **140** can perform other functions using other methods in addition to and/or aside from those shown in FIG. **14**. Referring to the description above with respect to FIGS. **1** through **13**, but using reference numbers from FIGS. **1** and **2** by way of example in this case, the method shown in the flowchart **1497** of FIG. **14** begins at the START step and proceeds to step **1461**, where a LCA **220** is inserted into a casing string **115** for a subsea wellbore **111**. The LCA **220** may include a receptacle **225** disposed toward its top end. The LCA **220** may include a LCA wall **224** that forms a LCA cavity **229**. The receptacle **225** may include a receptacle wall **226** that form a receptacle cavity **227**.

[0123] The LCA cavity **229** (which may include the receptacle cavity **227**) may have disposed therein toward its top end a first barrier **230-1** (e.g., as a result of performing step **1362** of FIG. **13** above). The first barrier **230-1** may form a first substantial fluidic seal with the LCA wall **224** (which may include the receptacle wall **226**). The first barrier **230-1** may be configured to open (e.g., break) the first substantial fluidic seal when a first differential force applied to the first barrier **230-1** within the LCA cavity **229** (which may include part of the receptacle cavity **227**) reaches a first threshold value. The LCA cavity **229** (which may include part of the receptacle cavity **227**) below the first barrier **230-1** may be filled with a working fluid **282** using, for example, the working fluid control system **170**. The LCA **220** may be inserted into the casing string **115** using the driving system **180**.

[0124] In step **1462**, the casing string **115** is filled with packer fluid **283**. The casing string may include a production packer **259** that limits the depth within the casing string **115** that the packer fluid **283** may fill. The packer fluid **283** may cover the top part of the LCA **220** in the casing string **115**, but the first barrier **230-1** keeps the packer fluid **283** from flowing into the LCA cavity **229**. The packer fluid **283** may be injected into the casing string **115** using a system that is equivalent to the working fluid control system **170**, but directed to use with the packer fluid **283**.

[0125] In step **1463**, the UCA **250** is inserted into the casing string **115**. The UCA **250** may include a tubing string **255** having a tubing string wall **254** that forms a tubing string cavity **257**. The tubing string cavity **257** may have disposed therein toward its bottom end a second barrier **230-2** (e.g., as a result of performing step **1364** of FIG. **13** above). The second barrier **230-2** may form a second substantial fluidic seal with the tubing string wall **254**. The second barrier **230-2** may be configured to open (e.g., break) the second substantial fluidic seal when a second differential force applied to the second barrier **230-2** within the tubing string cavity **257** reaches a second threshold value. The tubing string cavity **257** of the UCA **250** above the second barrier **230-2** may be filled with the working fluid **282** using, for example, the working fluid control system **170**. The UCA **250** may be inserted into the casing string **115** using the driving system **180**.

[0126] In step **1464**, the bottom end of the UCA **250** is engaged with the top end of the receptacle **225** of the LCA **220**. The bottom end of the UCA **250** may become engaged with the top end of the receptacle **225** of the LCA **220** using the driving system **180**. Specifically, the driving system **180** may lower the UCA **250** within the casing string **115** until the bottom end of the UCA **250** enters the receptacle cavity **227** of the receptacle **225** of the LCA **220**.

[0127] In certain example embodiments, engaging the bottom end of the tubing string **255** with the top end of the receptacle **225** raises the second differential force beyond the second threshold value, resulting in opening (e.g., breaking) the second barrier **230-2**. In other example embodiments, engaging the bottom end of the tubing string **115** with the top end of the receptacle **225** runs the bottom end of the tubing string **255** into the first barrier **230-1** at a first differential force that exceeds the first threshold value, resulting in opening (e.g., breaking) the first barrier **230-1**. In some cases, the bottom end of the tubing string **255** engages the top end of the receptacle **225** when a tubing hanger **266** of the UCA **250** is positioned within a seat of a Xmas tree **177**. In such cases, the Xmas tree **177** may be a horizontal tree.

[0128] In optional step **1467**, a measurement from a sensor device **160** may be obtained. For example, sensor device **160-1** and/or sensor device **160-2** may be used to measure a parameter (e.g., a magnetic field, proximity) associated with the bottom part of the UCA **250** engaging the receptacle **225** of the LCA **220**. The measurement may be obtained by a user **175**. In addition, or in the alternative, the measurement may be obtained by a controller **104**. In optional step **1468**, The bottom end of the UCA **250** is confirmed to be engaged with the receptacle of the LCA based on the measurement. The confirmation may be made by a user **175**. In addition, or in the alternative, the confirmation may be made by a controller **104**.

[0129] In step **1469**, the working fluid control system **170** may be activated. The working fluid control system **170** may be activated by a user **175**. In addition, or in the alternative, the working fluid control system **170** may be activated by a controller **104**. In some cases, activating the working fluid control system may include pumping the working fluid **282** at a progressively higher rate into the tubing string cavity **257**. As a result, a differential force applied to the second barrier **230-2** and/or the first barrier **230-1** may be sufficiently high to exceed the applicable threshold value needed to open (e.g., break) that barrier **230**.

[0130] In certain example embodiments, activating the working fluid control system **170** raises the first differential force applied to the first barrier **230-1** until the first differential force exceeds the first threshold value, resulting in the first barrier **230-1** opening (e.g., breaking). In other example embodiments, activating the working fluid control system **170** raises the second differential force applied to the second barrier **230-2** until the second differential force exceeds the second threshold value, resulting in the second barrier **230-2** opening (e.g., breaking).

[0131] In step **1471**, a crown plug (e.g., crown plug **356**, crown plug **856**) is installed. The one or more crown plugs may be installed on a Xmas tree **177**. The one or more crown plugs may be installed after the tubing hanger **266** is stabbed into the Xmas tree **177**. The crown plug may be installed using equipment controlled from a floating structure **103**. When step **1471** is complete, the process may proceed to the END step.

[0132] In some cases, when example embodiments are directed to a method of establishing a subsea completion system, the method may include inserting a first barrier within a top end of a LCA cavity formed by a LCA wall of the LCA, where first barrier forms a first fluidic seal with the LCA wall, where the first barrier is configured to open the first fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and where the LCA with the first barrier is configured to be subsequently inserted into a casing string. Such a method may also include inserting a second barrier toward a bottom end of a tubing string of a UCA, where the tubing string has a tubing string cavity formed by a tubing string wall, where the second barrier forms a second fluidic seal with the tubing string wall, where the second barrier is configured to open the second fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and where the UCA with the second barrier is configured to be inserted into the casing string after the LCA is inserted into the casing string.

[0133] In such cases, the method of establishing a subsea completion system may further include calibrating the first barrier to open the first fluidic seal at approximately the first threshold value, and calibrating the second barrier to open the second fluidic seal at approximately the second threshold value. In such cases, inserting the second barrier may include inverting a standard orientation of the second barrier when the second barrier comprises a sub, where the sub comprises a glass disc, a sliding sleeve, and a shear pin used to hold the sliding sleeve in place.

[0134] In some cases, when example embodiments are directed to a method of implementing a subsea completion system, the method may include inserting a LCA into a casing string for a subsea wellbore, where the LCA includes a receptacle disposed toward its top end and has a LCA wall that forms a LCA cavity, where the LCA cavity has disposed therein toward its top end a first barrier, where first barrier forms a first substantial fluidic seal with the LCA wall, where the first barrier is configured to open the first substantial fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and where the LCA below the first barrier is filled with a working fluid. Such a method may also include filling the casing string with a packer fluid, and inserting a UCA into the casing string, wherein the UCA includes a tubing string having a tubing string wall that forms a tubing string cavity, where the tubing string cavity has disposed therein toward its bottom end a second barrier, where the second barrier forms a second substantial fluidic seal with the tubing string wall, where the second barrier is configured to open the second substantial fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and where the UCA above the second barrier is filled with the working fluid. Such a method may further include engaging the bottom end of the UCA with the top end of the receptacle of the LCA, activating a working fluid control system, and installing a crown plug.

[0135] In such cases, the method of implementing a subsea completion system may also include, prior to activating the working fluid control system to raise the first differential force applied to the first barrier, obtaining a measurement from a sensor device positioned proximate to the bottom end of the UCA, and confirming, based on the measurement, that the bottom end of the UCA is engaged with the receptacle of the LCA. In addition, or in the alternative, in such cases, activating the working fluid control system may include pumping the working fluid at a progressively higher rate into the tubing string cavity. In addition, or in the alternative, in such cases, engaging the bottom end of the tubing string with the top end of the receptacle may raise the second differential force beyond the second threshold value. In addition, or in the alternative, in such cases, activating the working fluid control system may raise the first differential force applied to the first barrier until the first differential force exceeds the first threshold value.

[0136] In addition, or in the alternative, in such cases, engaging the bottom end of the tubing string with the top end of the receptacle may run the bottom end of the tubing string into the first barrier at a first differential force that exceeds the first threshold value. In addition, or in the alternative, in such cases, activating the working fluid control system may raise the second differential force applied to the second barrier until the second differential force exceeds the second threshold value. In addition, or in the alternative, in such cases, the bottom end of the tubing string may engage the top end of the receptacle when a tubing hanger of the UCA is positioned within a seat of a Xmas tree. In addition, or in the alternative, in such cases, the Xmas tree may be a horizontal Xmas tree.

[0137] Example embodiments can be used to safely and reliably install crown plugs in the tubing hanger. Example embodiments can use multiple barriers to isolate parts of a completion assembly so that little or no packer fluid is introduced to the subterranean formation. Example embodiments can be used for any of a number of field operations at various pressures, flow rates, and temperatures. Example embodiments can provide a number of benefits. Such benefits can include, but are not limited to, ease of use, reduction in costs, improved safety and reliability, reduced need of specialized equipment, configurability, time savings, and compliance with applicable industry standards and regulations.

[0138] Although embodiments described herein are made with reference to example embodiments, it should be appreciated by those skilled in the art that various modifications are well within the scope and spirit of this disclosure. Those skilled in the art will appreciate that the example embodiments described herein are not limited to any specifically discussed application and that the embodiments described herein are illustrative and not restrictive. From the description of the example embodiments, equivalents of the elements shown therein will suggest themselves to those skilled in the art, and ways of constructing other embodiments using the present disclosure will suggest themselves to practitioners of the art. Therefore, the scope of the example embodiments is not limited herein.

Claims

1. A subsea completion system comprising: a lower completion assembly (LCA) comprising a receptacle disposed at a top end of the LCA, wherein the LCA comprises a LCA wall that forms a LCA cavity; a first barrier disposed within the LCA cavity, wherein the first barrier forms a first fluidic seal with the LCA wall, and wherein the first barrier is configured to open the first fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value; an upper completion assembly (UCA) comprising a tubing string disposed toward a bottom end of the UCA, wherein the tubing string comprises a tubing string wall that forms a tubing string cavity; and a second barrier disposed within the tubing string cavity, wherein the second barrier forms a second fluidic seal with the tubing string wall, and wherein the second barrier is configured to open the second fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value.

2. The subsea completion system of claim 1, wherein the receptacle comprises a polished bore receptacle.

3. The subsea completion system of claim 1, wherein the LCA wall has a first diameter that is larger than a second diameter of the tubing string wall.

4. The subsea completion system of claim 1, wherein the second barrier comprises a glass disc.
5. The subsea completion system of claim 4, wherein the glass disc is part of a sub integrated with the tubing string.
6. The subsea completion system of claim 5, wherein the sub further comprises a shear pin that is configured to shear when the second differential force applied to the second barrier within the tubing string reaches the second threshold value.
7. The subsea completion system of claim 6, wherein the sub further comprises a piercing device that is configured to break the glass disc of the second barrier when the glass disc is forced upon the piercing device after the shear pin shears.
8. The subsea completion system of claim 6, wherein the sub further comprises a sliding sleeve configured to be held in place by the shear pin, wherein the sliding sleeve is configured to slide after the shear pin shears and allows the glass disc to move, wherein the glass disc is configured to break when glass disc moves into the piercing device, and wherein the second differential force reaches the second threshold value when the bottom end of the tubing string of the UCA is stabbed into the top end of the receptacle of the LCA.
9. The subsea completion system of claim 1, further comprising: a third barrier disposed within the LCA cavity, wherein the third barrier forms a third fluidic seal with the LCA wall, wherein the third barrier is configured to open the third fluidic seal when a third differential force applied to the third barrier within the LCA cavity reaches a third threshold value, and wherein the third barrier is positioned below the first barrier.
10. The subsea completion system of claim 1, wherein the first barrier comprises a ball valve.
11. The subsea completion system of claim 1, wherein the second threshold value is less than the first threshold value.
12. The subsea completion system of claim 1, wherein the UCA and the LCA are disposed within a casing string.
13. The subsea completion system of claim 12, wherein the first barrier prevents a packer fluid disposed between the LCA and the casing string from entering the LCA cavity when the first barrier forms the first fluidic seal with the LCA wall, and wherein the second barrier prevents the packer fluid disposed between the UCA and the casing string from entering the tubing string cavity when the second barrier forms the second fluidic seal with the tubing string wall.
14. The subsea completion system of claim 12, wherein the first barrier seals a working fluid within the LCA cavity below the first barrier when the first barrier forms the first fluidic seal with the LCA wall, and wherein the second barrier seals the working fluid within the tubing string cavity above the second barrier when the second barrier forms the second fluidic seal with the tubing string wall.
15. The subsea completion system of claim 14, further comprising: a working fluid control system that is configured to control a flow of the working fluid into the UCA from a top end of the UCA, wherein the working fluid system is further configured to raise the pressure through the working fluid within the tubing string cavity beyond the first threshold value, after the second barrier opens the second fluidic seal with the tubing string wall, to cause the first barrier to open the first fluidic seal with the LCA wall.
16. The subsea completion system of claim 1, further comprising: a driving system that is configured to move the UCA within the casing string, wherein the driving system is further configured to lower the bottom end of the tubing string of the UCA into the top end of the receptacle of the LCA.
17. The subsea completion system of claim 16, wherein the driving system is further configured to lower the LCA into place within the casing string prior to lowering the UCA toward the LCA within the casing string.
18. The subsea completion system of claim 16, further comprising: a sensor device that is configured to provide an indication when the bottom end of the tubing string of the UCA is stabbed

into the top end of the receptacle of the LCA, wherein the indication triggers the working fluid control system to raise the pressure through the working fluid within the tubing string cavity so that the first differential force imposed on the first barrier exceeds the first threshold value.

19. A method of establishing a subsea completion system, the method comprising: inserting a first barrier within a top end of a lower completion assembly (LCA) cavity formed by a LCA wall of the LCA, wherein first barrier forms a first fluidic seal with the LCA wall, wherein the first barrier is configured to open the first fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and wherein the LCA with the first barrier is configured to be subsequently inserted into a casing string; and inserting a second barrier toward a bottom end of a tubing string of an upper completion assembly (UCA), wherein the tubing string has a tubing string cavity formed by a tubing string wall, wherein the second barrier forms a second fluidic seal with the tubing string wall, wherein the second barrier is configured to open the second fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and wherein the UCA with the second barrier is configured to be inserted into the casing string after the LCA is inserted into the casing string.

20. A method of implementing a subsea completion system, the method comprising: inserting a lower completion assembly (LCA) into a casing string for a subsea wellbore, wherein the LCA comprises a receptacle disposed toward its top end and has a LCA wall that forms a LCA cavity, wherein the LCA cavity has disposed therein toward its top end a first barrier, wherein first barrier forms a first substantial fluidic seal with the LCA wall, wherein the first barrier is configured to open the first substantial fluidic seal when a first differential force applied to the first barrier within the LCA cavity reaches a first threshold value, and wherein the LCA below the first barrier is filled with a working fluid; filling the casing string with a packer fluid; inserting an upper completion assembly (UCA) into the casing string, wherein the UCA comprises a tubing string having a tubing string wall that forms a tubing string cavity, wherein the tubing string cavity has disposed therein toward its bottom end a second barrier, wherein the second barrier forms a second substantial fluidic seal with the tubing string wall, wherein the second barrier is configured to open the second substantial fluidic seal when a second differential force applied to the second barrier within the tubing string cavity reaches a second threshold value, and wherein the UCA above the second barrier is filled with the working fluid; engaging the bottom end of the UCA with the top end of the receptacle of the LCA; activating a working fluid control system; and installing a crown plug.